

Unifying Special Relativity and Quantum Mechanics

Lecture 12 — Internal Symmetries and the Reason for Antiparticles

(Wigner–Weyl Realization, $U(1)$ Charge, Charged Scalar and Dirac Fields)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–11: everything in the normal text and in the **yellow / pink** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones). Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, and connections to earlier lectures.

Pagination of this scan. The notes open a *new chapter* and restart the page numbering from 1 for the **sixth** time. To keep these sheets apart from all the earlier restarts they are marked [p. 1*****]–[p. 15*****], with **six stars**. There is no overlap with Lecture 11 this time: the scan begins cleanly at a new section heading, and page 15 is the last written sheet (page 16 of the PDF is blank).

A change of subject. Lectures 8–11 were about *space-time* symmetry: how the Poincaré group constrains fields and particles. This lecture turns to *internal* symmetry — a compact group G acting on the particle labels and commuting with all the Poincaré generators. The two topics meet in the middle of the lecture, where demanding that a field transform sensibly under G and under Lorentz forces the existence of antiparticles.

Where we left off. Lecture 11 closed the massless chapter: the four-vector field cannot carry the $SE(2)$ translations, so polarization vectors live in equivalence classes, which becomes gauge invariance $A^\mu \equiv A^\mu + \partial^\mu \alpha$ for the photon and $h^{\mu\nu} \equiv h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ for the graviton. None of that machinery is needed here; this lecture starts fresh.

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1 Symmetries realized à la Wigner–Weyl [p. 1****]

Consider the case in which we have, as symmetry group,

$$ISO^\uparrow(1,3) \otimes G$$

G is a **compact Lie group** with generators Q^A

We have, at the level of the algebra,

$$[P^\mu, Q^A] = [J^{\mu\nu}, Q^A] = 0$$

We call G an “*internal*” symmetry group

This symmetry is implemented via the following points:

- 1) We assume that the vacuum is left invariant by the action of G :

$$Q^A|0\rangle = 0$$

symmetry realized à la Wigner–Weyl

- 2) Transformations of G act on the physical states as unitary operators.

$\hookrightarrow$ Physical states will be described by states of the form ...

Added explanation (not in the original notes)

What “internal” means, and why the commutators matter. $[P^\mu, Q^A] = [J^{\mu\nu}, Q^A] = 0$ says the generators of G do not move points in spacetime and do not rotate or boost anything: they act only on extra, “internal” labels carried by the particle. Practically, this is what allows the one-particle Hilbert space to factorize as $\mathcal{H}_{\text{Poincaré}} \otimes \mathcal{H}_G$, which is exactly the tensor product written on the next page. It also means G cannot change mass or spin — so all the states in one G -multiplet are automatically degenerate in mass.

Why the direct product is not the only option (Coleman–Mandula). The notation $ISO^\uparrow(1,3) \otimes G$ is a *direct* product, not something more entangled. That is not a matter of convenience: the Coleman–Mandula theorem states that, under mild assumptions (non-trivial S -matrix, mass gap, finitely many species below any mass), the largest possible symmetry Lie algebra of the S -matrix *is* the direct sum of the Poincaré algebra and an internal algebra. Trying to mix them non-trivially fails. (The one loophole is graded algebras — supersymmetry — where the extra generators are fermionic.)

The two ways a symmetry can be “realized”. The assumption $Q^A|0\rangle = 0$ is *Wigner–Weyl* realization: the vacuum is invariant, the symmetry is manifest, and states organize into complete multiplets of G with equal masses. The alternative — $Q^A|0\rangle \neq 0$ — is the *Nambu–Goldstone* realization: the symmetry is spontaneously broken, no multiplet structure appears in the spectrum, and each broken generator produces a massless boson instead. This lecture treats only the first case; the label “à la Wigner–Weyl” in the section title is there precisely to mark the choice.

Why G compact. Compactness guarantees that the finite-dimensional unitary irreps are finite-dimensional and that the generators can be taken Hermitian with discrete spectra — which is what makes “charges” sensible quantum numbers.

2 Physical states and the example $G \equiv U(1)$ [p. 2****]

$$|M, \vec{p}, S, \lambda\rangle \otimes |\alpha\rangle \equiv |\alpha, M, \vec{p}, S, \lambda\rangle$$

$\hookrightarrow$ “ α ” are additional labels that specify how the physical states transform under G .

Notable example: $G \equiv U(1)$

The simplest example is that of $G \equiv U(1)$.

- We have one generator, “ Q ”.
- We consider the eigenvalues $Q|q\rangle = q|q\rangle$ where, since Q must be Hermitian, $q \in \mathbb{R}$.
- Group elements are given by the exponential map:

$$U(\theta) \equiv \exp(-i\theta Q)$$

$$U(\theta)|q\rangle = e^{-i\theta q}|q\rangle, \quad q \in \mathbb{Z}$$

The “*quantization condition*” $q \in \mathbb{Z}$ comes from the condition $U(\theta) = U(\theta + 2\pi)$

► Irreps of $U(1)$ are labelled by a fixed value of $q \in \mathbb{Z}$. Therefore, if we associate fundamental ...

Added explanation (not in the original notes)

Reading the tensor product. The first factor carries everything the Poincaré group knows about: mass M , momentum $\vec{p}$, spin S , and the spin projection / helicity λ . The second factor carries the internal labels. Because the two groups commute, a G transformation acts only on $|\alpha\rangle$ and a Poincaré transformation only on the first factor — neither disturbs the other.

Where $q \in \mathbb{Z}$ comes from. $U(1)$ as a manifold is a circle: the parameter θ and $\theta + 2\pi$ are the *same* group element. So a genuine representation must satisfy $U(\theta) = U(\theta + 2\pi)$, i.e. $e^{-iq\theta} = e^{-iq(\theta+2\pi)}$, which needs $e^{-2\pi iq} = 1$ and hence q integer. This is the same argument that quantizes the orbital angular momentum quantum number m in elementary quantum mechanics.

Why one generator. $U(1)$ is one-dimensional as a Lie group, and abelian, so all its irreducible unitary representations are one-dimensional: multiplication by a phase. The whole content of an irrep is the single number q — the *charge*.

3 One-particle states with charge; the projective subtlety [p. 3****]

... particles to (unitary) irreps of the underlying symmetry group, we can say that physical one-particle states are characterized by

$$\left\{ |M, \vec{p}, S, \lambda\rangle \otimes |q\rangle \equiv |q, M, \vec{p}, S, \lambda\rangle \right\} \quad M, S, q \text{ fixed}$$

$\hookrightarrow$ These states with M, S, q fixed describe a particle with mass “ M ”, spin “ S ” and **charge** “ q ”.

Comment

The condition $q \in \mathbb{Z}$ is not strict in a QM setting. Consider a generic value $q \in \mathbb{R}$ and combine two transformations:

$$U(\theta_1)U(\theta_2) = e^{-iq(\theta_1+\theta_2)} = U(\theta_1 + \theta_2)$$

Take now θ_1 and θ_2 such that $\theta_1 + \theta_2 = 2\pi$:

$$U(\theta_1)U(\theta_2) = e^{-iq2\pi} = U(2\pi) \stackrel{!}{=} 1$$

$\hookrightarrow U(1)$ is defined by the “global” condition
 $U(2\pi) = 1$

Whatever $q \in \mathbb{R}$, we can always write it as $q = [q] + r$, with $[q]$ integer part and “ r ” the fractional part, $r \in [0, 1)$.

Added explanation (not in the original notes)

What is being probed here. In ordinary group theory one demands $U(2\pi) = \mathbf{1}$ exactly. In quantum mechanics, states are rays: $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ are the *same* physical state. So the honest requirement is only $U(2\pi) = e^{i\varphi}\mathbf{1}$ — equality up to a phase. That weaker requirement is what a *projective* representation satisfies, and it is what the next page exploits.

This is the same phenomenon as spin-1/2. A 2π rotation acts on a spinor as -1 , not $+1$: the rotation group $SO(3)$ is represented only projectively, and one passes to the covering group $SU(2)$ to make it a genuine representation. The $U(1)$ story on the next page is the exact analogue, with covering group $(\mathbb{R}, +)$ instead of $SU(2)$.

4 Projective representations and how fields transform [p. 4****]

We thus get

$$U(\theta_1)U(\theta_2) = e^{-i([q]+r)2\pi} = \underbrace{e^{-i[q]2\pi}}_1 \underbrace{e^{-ir2\pi}}_{\substack{\text{since } r \in [0,1) \\ \text{this is in general} \\ \text{a non-vanishing phase}}} = e^{-ir2\pi} \neq 1$$

Consequently, irreps with $q \in \mathbb{R}$ are not ordinary representations of $U(1)$ but **projective representations**. In a QM setting, we should include them so, in principle, q could be any real number.

$\hookrightarrow$ Group theory side, this is because $U(1)$ is not, as a manifold, simply connected and its universal covering group is precisely $(\mathbb{R}, +)$.

3) We need to specify how the fields transform under the symmetry group G .

$$\Phi^a(x) \xrightarrow{\vec{\theta}} \Phi'^a(x) = U(\vec{\theta})^\dagger \Phi^a(x) U(\vec{\theta}) = (D(\vec{\theta}))^a_b \Phi^b(x)$$

$\vec{\theta}$ is a set of $\dim G$ parameters that identify a group transformation $U(\vec{\theta}) \equiv \exp(-i\theta^A Q^A)$ $D(\vec{\theta}) = \exp(-i\theta^A T^A)$

Added explanation (not in the original notes)

Why the integer part drops out. $e^{-i[q]2\pi} = 1$ because $[q]$ is an integer; only the fractional part r survives, producing a leftover phase $e^{-2\pi ir}$. That phase never affects any probability — which is exactly why quantum mechanics tolerates it.

“Not simply connected”, unpacked. A loop that winds once around the circle $U(1)$ cannot be contracted to a point without leaving the circle. Its universal cover — the simply connected group that maps onto it — is the real line $(\mathbb{R}, +)$, wrapped onto the circle by $\theta \mapsto e^{i\theta}$. Representations of $(\mathbb{R}, +)$ are labelled by *any* real q ; only those with $q \in \mathbb{Z}$ descend to genuine representations of $U(1)$ itself. The rest are the projective ones.

A physical footnote. Whether electric charge is quantized in nature is therefore *not* settled by this argument alone: for a pure $U(1)$ nothing forbids irrational charges. Quantization does follow if the $U(1)$ sits inside a larger compact non-abelian group (as in grand unification), or in the presence of a magnetic monopole (Dirac’s argument). Experimentally, charges are quantized to extraordinary precision, which is usually read as a hint for one of these.

Two different representations, one symmetry. Note carefully that *two* sets of generators appear: Q^A in $U(\vec{\theta})$ and T^A in $D(\vec{\theta})$. They represent the same abstract group elements, but act on different spaces — states and field components respectively. The indices a, b on Φ^a are internal (not Lorentz) indices, running over the components of the G multiplet the field belongs to. The next page relates the two.

5 The infinitesimal relation between Q^A and T^A [p. 5*****]

- T^A are the generators of G in the representation according to which fields transform.
- Q^A are the generators of G in the representation according to which physical states transform.

At the infinitesimal level, we write (at the first order in $\vec{\theta}$):

$$\begin{aligned} U(\vec{\theta})^\dagger \Phi^a(x) U(\vec{\theta}) &= (D(\vec{\theta}))^a_b \Phi^b(x) \\ &\simeq \left(\delta^a_b - i\theta^A (T^A)^a_b \right) \Phi^b(x) \\ &= \Phi^a(x) - i\theta^A (T^A)^a_b \Phi^b(x) \\ (\mathbf{1} + i\theta^A Q^A) \Phi^a(x) (\mathbf{1} - i\theta^A Q^A) &= \Phi^a(x) + i\theta^A [Q^A, \Phi^a(x)] \end{aligned}$$

From the comparison, we get

$$[Q^A, \Phi^a(x)] = -(T^A)^a_b \Phi^b(x)$$

Added explanation (not in the original notes)

Doing the comparison carefully. Expanding the left-hand side, $U^\dagger = e^{+i\theta^A Q^A} \simeq \mathbf{1} + i\theta^A Q^A$ and $U \simeq \mathbf{1} - i\theta^A Q^A$, so

$$U^\dagger \Phi^a U \simeq \Phi^a + i\theta^A (Q^A \Phi^a - \Phi^a Q^A) = \Phi^a + i\theta^A [Q^A, \Phi^a].$$

Matching the $O(\theta)$ terms with $-i\theta^A (T^A)^a_b \Phi^b$ and dividing by $i\theta^A$ gives the boxed result. The relative minus sign is not a convention one may drop — it is what makes the two representations consistent.

What the boxed relation is for. It converts a statement about representations into a statement about operators. Once you know which multiplet of G a field sits in (i.e. you know T^A), the commutator of the field with the conserved charge is completely fixed. Conversely, the commutator can be used to *read off* the charge of a field — which is exactly how it will be used on [p. 10*****], where $[Q, \Phi^A(x)] = -q\Phi^A(x)$ tells us the field carries charge q .

Consistency check. Applying the Jacobi identity to $[Q^A, [Q^B, \Phi]] - [Q^B, [Q^A, \Phi]] = [[Q^A, Q^B], \Phi]$ and using the boxed relation shows that the T^A satisfy the *same* Lie algebra as the Q^A , $[T^A, T^B] = i f^{ABC} T^C$ — as they must, being two representations of one group.

6 The reason for antiparticles [p. 5*****–6*****]

Let us discuss the case of $G \equiv U(1)$ further. We consider the possibility of having particles with some charge “ q ”. The physical states describing such particles are

$$|q, m, \vec{p}, \lambda\rangle = \sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)|0\rangle$$

index for the charge q

index with informations about
mass, spin

creation operators do have the
same additional label “ q ”

■ We now derive the transformation rules for creation and annihilation operators under G :

$$U(\theta)|q, m, \vec{p}, \lambda\rangle = e^{-i\theta Q}|q, m, \vec{p}, \lambda\rangle = e^{-i\theta q}|q, m, \vec{p}, \lambda\rangle$$

On the left-hand side we write

$$\begin{aligned} U(\theta)\sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)|0\rangle &= U(\theta)\sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)U(\theta)^{-1}U(\theta)|0\rangle \\ &= \sqrt{2E_p} U(\theta)a^\dagger(q, m, \vec{p}, \lambda)U(\theta)^{-1}|0\rangle, \end{aligned}$$

while on the right-hand side: $e^{-i\theta q}\sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)|0\rangle$.

From the comparison:

$$\begin{aligned} U(\theta) a^\dagger(q, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{-i\theta q} a^\dagger(q, m, \vec{p}, \lambda) \\ U(\theta) a(q, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{+i\theta q} a(q, m, \vec{p}, \lambda) \end{aligned}$$

Added explanation (not in the original notes)

The one step that makes the derivation work is inserting $\mathbf{1} = U(\theta)^{-1}U(\theta)$ between $a^\dagger$ and $|0\rangle$, then using $U(\theta)|0\rangle = |0\rangle$ — which is precisely assumption 1) of [p. 1*****], the Wigner–Weyl condition $Q|0\rangle = 0$. Without an invariant vacuum this whole derivation collapses.

Why the second line follows from the first. Take the Hermitian conjugate of the first relation. Since U is unitary, $(Ua^\dagger U^{-1})^\dagger = UaU^{-1}$, and $(e^{-i\theta q})^* = e^{+i\theta q}$ for real θ, q . Physically: $a^\dagger$ adds one unit of charge q to a state and a removes it, so they must rotate with opposite phases.

The factor $\sqrt{2E_p}$ is the relativistic normalization convention used throughout the course; it plays no role in this argument and simply passes through.

7 The particle–field connection under $U(1)$ [p. 7*****]

We now investigate the particle–field connection. Let us consider the quantum field

$$\Phi^A(x) = \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} a(q, m, \vec{p}, \lambda) + v^A(q, m, \vec{p}, \lambda) e^{ip \cdot x} a^\dagger(q, m, \vec{p}, \lambda) \right]$$

with

$$U(\Lambda, a) \Phi^A(x) U(\Lambda, a)^\dagger = (D(\Lambda^{-1}))^A_B \Phi^B(\Lambda x + a)$$

↔ We consider a quantum field with generic transformation properties under Lorentz. Consequently, as we have discussed, the coefficient functions must obey the particle–field connection as far as space-time symmetries are concerned. Furthermore, creation and annihilation operators are combined in such a way to preserve microcausality.

In addition, the field must transform under $U(1)$ in a way that is compatible with creation and annihilation operators. Let us investigate this point more carefully.

Added explanation (not in the original notes)

Reading the notation. The index A is a *Lorentz* index here (it labels the components of whatever Lorentz representation the field sits in — scalar, vector, spinor), while the arguments $q, m, \vec{p}, \lambda$ are the particle labels: charge, mass, momentum, spin projection. The two requirements now in play are logically independent:

- (i) *Lorentz*: the coefficient functions u^A, v^A must intertwine the Lorentz representation with the little-group representation — everything Lectures 8–11 were about.
- (ii) *Internal*: the same field must also transform simply under $U(1)$.

The next page shows that (ii) cannot be satisfied with the field as written, and that repairing it requires a genuinely new physical ingredient.

Why microcausality forced this particular pairing. From Lectures 6–9: a field containing *only* a (or only $a^\dagger$) has an equal-time bracket at spacelike separation that does not vanish. Combining the two pieces with the specific relative weight produces the cancellation $\Delta_+(x - y) \mp \Delta_+(y - x) = 0$ outside the light cone. So the presence of *both* terms is not optional — and that is exactly what will collide with the $U(1)$ requirement.

8 The clash, and the postulate of antiparticles [p. 8*****]

$$\begin{aligned} U(\theta) \Phi^A(x) U(\theta)^{-1} &= \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} U(\theta) a(q, m, \vec{p}, \lambda) U(\theta)^{-1} \right. \\ &\quad \left. + v^A(q, m, \vec{p}, \lambda) e^{ip \cdot x} U(\theta) a^\dagger(q, m, \vec{p}, \lambda) U(\theta)^{-1} \right] \\ &= \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} e^{+i\theta q} a(q, m, \vec{p}, \lambda) \right. \\ &\quad \left. + v^A(q, m, \vec{p}, \lambda) e^{ip \cdot x} e^{-i\theta q} a^\dagger(q, m, \vec{p}, \lambda) \right] \end{aligned}$$

► We have a problem here because it is not possible to bring the transformed field in a form of the kind

$$U(\theta) \Phi^A(x) U(\theta)^{-1} = e^{i\theta q} \Phi^A(x)$$

in which the right-hand side is proportional to the field itself.

The **only** way out is to postulate, for each particle with charge “ q ”, the existence of a particle with the same labels as far as space-time symmetries are concerned (**same mass, spin**) but **opposite charge**, $\bar{q} = -q$.

Let us then repeat the previous construction under this assumption.

Added explanation (not in the original notes)

The clash in one line. The two terms of the field rotate with *opposite* phases: $e^{+i\theta q}$ multiplies the a piece, $e^{-i\theta q}$ the $a^\dagger$ piece. A single overall factor can therefore never be pulled out — unless $q = 0$, in which case there is no charge to speak of. So a charged field built out of one species is impossible.

Why the field must transform by a single phase. If it did not, the transformed field would be a *different* operator, not a multiple of Φ^A , and the relation $[Q, \Phi^a] = -(T^a)^a_b \Phi^b$ of [p. 5*****] could not hold for any T . Concretely: one could not write down a $U(1)$ -invariant Lagrangian, because terms like $\Phi^\dagger \Phi$ would fail to be invariant. Simple transformation of the field is not an aesthetic preference; it is what makes an interacting theory with this symmetry possible at all.

The resolution, stated plainly. Keep the a piece as the annihilator of a charge- q particle, but replace the $a^\dagger$ piece by the *creator of a different particle* carrying charge $-q$. Then both pieces *lower* the total charge by q , both rotate with the same phase, and the field transforms simply. That new particle is the **antiparticle**.

This is a genuine prediction, and it is worth pausing on. Nothing was assumed about antimatter. The chain was: internal symmetry $\Rightarrow$ charged particles $\Rightarrow$ microcausality demands both a and $a^\dagger$ in the field $\Rightarrow$ the two phases clash $\Rightarrow$ a second species with opposite charge, same mass, same spin *must* exist. Note the contrast with Lecture 9: there, causality gave $|\kappa| = |\eta|$ — particle and antiparticle appear with equal weight — but did not by itself require them to be *distinct*. Here the internal charge is what makes them distinct. A neutral particle ($q = 0$) has no clash and can be its own antiparticle, which is why the photon and the Z^0 are self-conjugate while the $W^\pm$ and the electron are not.

9 Two species of creation and annihilation operators [p. 9*****]

■ Creation and annihilation operators

$$\begin{aligned} |q, m, \vec{p}, \lambda\rangle &= \sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda) |0\rangle \\ |\bar{q}, m, \vec{p}, \lambda\rangle &= \sqrt{2E_p} b^\dagger(\bar{q}, m, \vec{p}, \lambda) |0\rangle \end{aligned}$$

We introduce two distinct sets of creation and annihilation operators, each of which with its own commutators / anticommutation rules:

$$\begin{cases} \left[a(m, q, \vec{p}, \lambda), a^\dagger(m, q, \vec{q}', \lambda') \right]_\pm = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}') \delta_{\lambda\lambda'} \\ [a, a]_\pm = [a^\dagger, a^\dagger]_\pm = 0 \end{cases} \quad \begin{cases} \left[b(m, \bar{q}, \vec{p}, \lambda), b^\dagger(m, \bar{q}, \vec{k}', \lambda') \right]_\pm = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}') \delta_{\lambda\lambda'} \\ [b, b]_\pm = [b^\dagger, b^\dagger]_\pm = 0 \end{cases}$$

► In addition, all commutators / anticommutators between $a, a^\dagger$ and $b, b^\dagger$ vanish since we talk about different particles.

Under $U(1)$, we have the following transformation properties:

$$\begin{aligned} U(\theta) a^\dagger(q, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{-i\theta q} a^\dagger(q, m, \vec{p}, \lambda) \\ U(\theta) a(q, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{+i\theta q} a(q, m, \vec{p}, \lambda) \end{aligned}$$

Added explanation (not in the original notes)

Why the two species commute with each other. Operators referring to genuinely different particles act on independent factors of the Fock space, so all their mutual brackets vanish. This is not an extra assumption about dynamics — it is the definition of “two different species”.

The $\pm$ notation. The subscript $\pm$ keeps both options open: commutators for bosons, anticommutators for fermions. Following the pattern of Lectures 7–9, statistics is not assumed here; it is fixed later by microcausality (and, as page 15***** records, the Dirac field ends up needing anticommutators). The important point on the next few pages is that *none* of the charge algebra depends on the choice — explicitly noted on [p. 12*****].

Transcription note. The argument lists are written slightly differently from place to place in the handwriting — $a(m, q, \vec{p}, \lambda)$ here versus $a(q, m, \vec{p}, \lambda)$ elsewhere, and the dummy momentum is called $\vec{q}'$ for the a 's but $\vec{k}$ for the b 's. These are the same objects; only the ordering of the labels and the name of the integration variable differ. I have kept the original letters so the transcription stays faithful.

10 The charged field, at last [p. 10*****]

$$\begin{aligned} U(\theta) b^\dagger(\vec{q}, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{+i\theta q} b^\dagger(\vec{q}, m, \vec{p}, \lambda) \\ U(\theta) b(\vec{q}, m, \vec{p}, \lambda) U(\theta)^{-1} &= e^{-i\theta q} b(\vec{q}, m, \vec{p}, \lambda) \end{aligned}$$

■ Quantum fields

We see that $a(q, m, \vec{p}, \lambda)$ has the same transformation properties of $b^\dagger(\vec{q}, m, \vec{p}, \lambda)$. We thus write:

$$\Phi^A(x) = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} a(q, m, \vec{p}, \lambda) + v^A(\vec{q}, m, \vec{p}, \lambda) e^{ip \cdot x} b^\dagger(\vec{q}, m, \vec{p}, \lambda) \right]$$

And we now have

$$\begin{aligned} U(\theta) \Phi^A(x) U(\theta)^{-1} &= e^{+i\theta q} \Phi^A(x) \\ [Q, \Phi^A(x)] &= -q \Phi^A(x) \end{aligned}$$

that now has the correct transformation rules.

Comments

1) The field $\Phi^A(x)$ is not Hermitian.

Added explanation (not in the original notes)

Why $b^\dagger$ carries the phase $e^{+i\theta q}$. The antiparticle has charge $\bar{q} = -q$. Applying the rule of the previous page with $q \rightarrow \bar{q} = -q$ gives $Ub^\dagger U^{-1} = e^{-i\theta\bar{q}}b^\dagger = e^{+i\theta q}b^\dagger$ — the same phase as a . That coincidence is the whole point: it is what lets a single factor $e^{i\theta q}$ be pulled out of the field.

Reading the field physically. Both terms *lower* the total charge by q : the first by destroying a particle of charge $+q$, the second by creating an antiparticle of charge $-q$. A charged field has a single, unambiguous action on charge — and its conjugate $\Phi^{A\dagger}$ raises it by q . This is what makes terms like $\Phi^\dagger\Phi$ invariant and $\Phi\Phi$ not.

Consistency with [p. 5***].** The general relation was $[Q^A, \Phi^a] = -(T^A)^a_b \Phi^b$. For $U(1)$ acting on a field of charge q the representation matrix is just the number $T = q$, so $[Q, \Phi^A] = -q\Phi^A$ — exactly the boxed result, now derived independently from the mode expansion. The two routes agreeing is a genuine check on the construction.

Why non-Hermiticity is the expected outcome, not a defect. A Hermitian field would satisfy $\Phi^\dagger = \Phi$ and hence both raise and lower charge — impossible unless $q = 0$. Compare the neutral fields of the earlier lectures: the real scalar (Lecture 6) and the real vector field V^μ of Lecture 10, both Hermitian, both describing particles that are their own antiparticles. Charge and Hermiticity are mutually exclusive.

11 The Hamiltonian and the charge operator [p. 11*****]

2) We can construct the Hamiltonian of the theory. In general, we have

$$H_0 = \sum_m \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3} E_p a^\dagger(m, \vec{p}, \lambda) a(m, \vec{p}, \lambda)$$

in this case, we sum over particle and anti-particle species

$$H_0 = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3} E_p \left[a^\dagger(q, m, \vec{p}, \lambda) a(q, m, \vec{p}, \lambda) + b^\dagger(\bar{q}, m, \vec{p}, \lambda) b(\bar{q}, m, \vec{p}, \lambda) \right]$$

3) We need a charge operator Q such that

$$\begin{aligned} Q|0\rangle &= 0 \\ Q|q, m, \vec{p}, \lambda\rangle &= q|q, m, \vec{p}, \lambda\rangle \\ [Q, \Phi^A(x)] &= -q\Phi^A(x) \end{aligned}$$

In a free theory, an educated guess is

$$Q = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3} \left[q a^\dagger(q, m, \vec{p}, \lambda) a(q, m, \vec{p}, \lambda) + \bar{q} b^\dagger(\bar{q}, m, \vec{p}, \lambda) b(\bar{q}, m, \vec{p}, \lambda) \right]$$

Added explanation (not in the original notes)

Transcription note. In the handwritten charge operator the b operators are written with the label q rather than $\bar{q}$, while their coefficient is correctly $\bar{q}$. Since the b 's were defined on [p. 9*****] as carrying $\bar{q}$, the labels have been made uniform above. This is a bookkeeping slip only — the coefficient, which is what matters, is right in the original.

Reading both operators the same way. $a^\dagger a$ and $b^\dagger b$ are number operators counting particles and antiparticles of momentum $\vec{p}$ and helicity λ . So

$$H_0 = \sum \int E_p (N_a + N_b), \quad Q = \sum \int (q N_a + \bar{q} N_b) = q \sum \int (N_a - N_b),$$

using $\bar{q} = -q$. The energy adds — both species cost E_p — while the charge *subtracts*: Q counts the net excess of particles over antiparticles. That is why Q is conserved in processes that create particle–antiparticle pairs while H_0 is not sensitive to the distinction.

Why $Q|0\rangle = 0$ is automatic here. Both terms are normal-ordered, with annihilation operators standing to the right, so each kills the vacuum. Note that no normal-ordering *constant* was subtracted — unlike H_0 , where the analogous reordering leaves the (divergent) zero-point energy. For the charge the two species' zero-point contributions come with opposite signs q and $\bar{q}$ and cancel, so the vacuum is exactly neutral.

Why “educated guess”. In a genuinely interacting theory Q would be obtained from Noether's theorem as $Q = \int d^3x j^0(x)$ with $\partial_\mu j^\mu = 0$. Here the notes take the free-field shortcut: write the natural candidate and then *verify* the three defining properties. That verification is the content of the next three pages.

12 Verifying the charge operator [p. 12*****–13*****]

Let us check.

1. We clearly have $Q|0\rangle = 0$.

2. Let us compute $[Q, a^\dagger(q, m, \vec{p}, \lambda)]$ and $[Q, b^\dagger(\bar{q}, m, \vec{p}, \lambda)]$.

► We do it without any assumption on the statistics of creation and annihilation operators.

$$\begin{aligned} [Q, a^\dagger(q, m, \vec{p}, \lambda)] &= \sum_{\lambda'} \int \frac{d^3\vec{k}}{(2\pi)^3} [q a^\dagger(q, m, \vec{k}, \lambda') a(q, m, \vec{k}, \lambda'), a^\dagger(q, m, \vec{p}, \lambda)] \\ &\quad \text{using } [AB, C] = A[B, C]_\pm \mp [A, C]_\pm B \\ &= \sum_{\lambda'} \int \frac{d^3\vec{k}}{(2\pi)^3} q a^\dagger(q, m, \vec{k}, \lambda') \underbrace{[a(q, m, \vec{k}, \lambda'), a^\dagger(q, m, \vec{p}, \lambda)]_\pm}_{(2\pi)^3 \delta_{\lambda\lambda'} \delta^{(3)}(\vec{k} - \vec{p})} \\ &= q a^\dagger(q, m, \vec{p}, \lambda) \end{aligned}$$

$$[Q, a^\dagger(q, m, \vec{p}, \lambda)] = q a^\dagger(q, m, \vec{p}, \lambda)$$

Similarly,

$$[Q, b^\dagger(\bar{q}, m, \vec{p}, \lambda)] = \sum_{\lambda'} \int \frac{d^3\vec{k}}{(2\pi)^3} [\bar{q} b^\dagger(\bar{q}, m, \vec{k}, \lambda') b(\bar{q}, m, \vec{k}, \lambda'), b^\dagger(\bar{q}, m, \vec{p}, \lambda)]$$

$$= \bar{q} b^\dagger(\bar{q}, m, \vec{p}, \lambda) = -q b^\dagger(\bar{q}, m, \vec{p}, \lambda)$$

$$\left[Q, b^\dagger(\bar{q}, m, \vec{p}, \lambda) \right] = -q b^\dagger(\bar{q}, m, \vec{p}, \lambda)$$

Added explanation (not in the original notes)

Why the result is statistics-independent — the key algebraic point. The identity used is

$$[AB, C] = A[B, C]_\pm \mp [A, C]_\pm B,$$

valid for either sign. With $A = a^\dagger(\vec{k})$, $B = a(\vec{k})$, $C = a^\dagger(\vec{p})$, the second term contains $[a^\dagger, a^\dagger]_\pm = 0$ — which vanishes for *both* choices of statistics — and the first term contains the canonical bracket, whose value $(2\pi)^3 \delta^{(3)} \delta_{\lambda\lambda'}$ is the same in both cases. So the two possibilities give literally the same answer. This is the precise sense of the pink annotation on the page: the charge algebra is blind to spin-statistics.

Checking the identity itself. For the upper (commutator) sign: $A[B, C]_+ - [A, C]_+ B = A(BC + CB) - (AC + CA)B = ABC - CAB = [AB, C]$. The lower sign works the same way with the roles of $+$ and $-$ exchanged. It is the standard trick for commuting a bilinear with a single operator.

Interpretation. $[Q, a^\dagger] = +q a^\dagger$ says $a^\dagger$ raises the eigenvalue of Q by q : acting on a state of charge Q_0 , one gets $Q a^\dagger|\psi\rangle = (Q_0 + q)a^\dagger|\psi\rangle$. Similarly $b^\dagger$ lowers it by q . These are exactly ladder relations, structurally identical to $[H_0, a^\dagger] = E_p a^\dagger$ for energy.

We thus have

$$\begin{aligned} Q|q, m, \vec{p}, \lambda\rangle &= Q\sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)|0\rangle \\ &= \sqrt{2E_p} \left(a^\dagger(q, m, \vec{p}, \lambda) \underbrace{Q}_{Q|0\rangle=0} + q a^\dagger(q, m, \vec{p}, \lambda) \right) |0\rangle \\ &= q\sqrt{2E_p} a^\dagger(q, m, \vec{p}, \lambda)|0\rangle = q|q, m, \vec{p}, \lambda\rangle \end{aligned}$$

and, similarly,

$$Q|\bar{q}, m, \vec{p}, \lambda\rangle = \bar{q}|\bar{q}, m, \vec{p}, \lambda\rangle = -q|\bar{q}, m, \vec{p}, \lambda\rangle$$

3. Finally, we compute $[Q, \Phi^A(x)]$:

$$\begin{aligned} [Q, \Phi^A(x)] &= \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[Q, u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} a(q, m, \vec{p}, \lambda) \right. \\ &\quad \left. + v^A(\bar{q}, m, \vec{p}, \lambda) e^{ip \cdot x} b^\dagger(\bar{q}, m, \vec{p}, \lambda) \right] \\ &= \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left\{ u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} [Q, a(q, m, \vec{p}, \lambda)] \right. \\ &\quad \left. + v^A(\bar{q}, m, \vec{p}, \lambda) e^{ip \cdot x} [Q, b^\dagger(\bar{q}, m, \vec{p}, \lambda)] \right\} \end{aligned}$$

13 The commutator with the field; two examples [p. 14*****-15*****]

Since

$$[Q, a^\dagger(q, m, \vec{p}, \lambda)] = q a^\dagger(q, m, \vec{p}, \lambda)$$

↓ taking the Hermitian conjugate

$$\left[a(q, m, \vec{p}, \lambda), Q \right] = q a(q, m, \vec{p}, \lambda) \quad \text{that is} \quad \left[Q, a(q, m, \vec{p}, \lambda) \right] = -q a(q, m, \vec{p}, \lambda)$$

we get

$$\begin{aligned} [Q, \Phi^A(x)] &= \sum_{\lambda} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[-q u^A(q, m, \vec{p}, \lambda) e^{-ip \cdot x} a(q, m, \vec{p}, \lambda) \right. \\ &\quad \left. - q v^A(\bar{q}, m, \vec{p}, \lambda) e^{ip \cdot x} b^\dagger(\bar{q}, m, \vec{p}, \lambda) \right] \\ &= -q \Phi^A(x). \end{aligned}$$

Added explanation (not in the original notes)

Watch the Hermitian conjugation. Q is Hermitian, so $[Q, a^\dagger]^\dagger = (Qa^\dagger - a^\dagger Q)^\dagger = aQ - Qa = [a, Q]$, and the right-hand side conjugates to qa (with q real). Hence $[a, Q] = qa$, i.e. $[Q, a] = -qa$. The sign flip is easy to lose and it is exactly what makes the two terms of the field come out with the *same* coefficient $-q$.

Why both terms share the coefficient $-q$. The first term carries $[Q, a] = -qa$; the second carries $[Q, b^\dagger] = \bar{q}b^\dagger = -qb^\dagger$. Different operators, different physical processes — destroying a particle versus creating an antiparticle — but identical effect on the charge. That common factor is what allows $-q$ to be pulled out, reproducing the boxed relation of [p. 10****] and closing the consistency loop.

All three defining properties are now verified, so the “educated guess” of [p. 11****] was correct: it really is the charge operator.

Example 1: the charged scalar field

$$\phi(x) = \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[e^{-ip \cdot x} a(q, \vec{p}) + e^{ip \cdot x} b^\dagger(\bar{q}, \vec{p}) \right]$$

► No transformation properties under Lorentz. It describes scalar (spin-0) particles and antiparticles.

Example 2: the Dirac field

The Dirac field is given by

$$\Psi^A(x) = \sum_{\sigma=\pm 1/2} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(q, \vec{p}, \sigma) e^{-ip \cdot x} a(q, \vec{p}, \sigma) + v^A(\bar{q}, \vec{p}, \sigma) e^{ip \cdot x} b^\dagger(\bar{q}, \vec{p}, \sigma) \right]$$

↔ It transforms non trivially under Lorentz.

$$\Psi(x) \xrightarrow{\Lambda} \exp \left(-\frac{i}{2} \omega_{\mu\nu} J_S^{\mu\nu} \right) \Psi(\Lambda^{-1}x)$$

$$J_S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu]$$

$\hookrightarrow$ Generators of Lorentz in the representation $\mathcal{D}^{(1/2,0)} \oplus \mathcal{D}^{(0,1/2)}$.

$\hookrightarrow$ It describes spin- $\frac{1}{2}$ massive particles with charge $+q$ and spin- $\frac{1}{2}$ antiparticles with charge $-q$.

The Dirac field verifies microcausality conditions only if anticommutation relations are involved.

Added explanation (not in the original notes)

The charged scalar, in context. This is the simplest possible illustration: no Lorentz indices at all, so $u^A = v^A = 1$ and the entire content of the field is the internal charge structure. Compare with the *real* scalar field of Lecture 6, which had a and $a^\dagger$ of the same species and was Hermitian: that one describes a neutral particle that is its own antiparticle. Adding a $U(1)$ charge is precisely what splits the single real field into a complex one with two species. Equivalently, $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$ built from two real fields of equal mass — which is why a charged scalar has *two* real degrees of freedom per momentum mode.

The Dirac field brings the whole course together. It carries, simultaneously:

- *Space-time structure* — the $(1/2, 0) \oplus (0, 1/2)$ representation, with the explicit generators $J_S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ (often written $\sigma^{\mu\nu}/2$). The direct sum of the two Weyl representations, rather than either alone, is what makes the field parity-covariant, exactly as $(1, 0) \oplus (0, 1)$ was for the field strength in Lecture 11.
- *Internal structure* — the charge q , with the antiparticle at $-q$: the electron and the positron.
- *Statistics* — fixed by microcausality, as in Lectures 7–9, but now landing on *anticommutators*. Recall the mechanism from Lecture 10: for integer spin, the spin sum is even under $p \rightarrow -p$ and only the commutator can kill the bracket outside the light cone; for half-integer spin the sum $\sum_s u\bar{u} = p_\mu \gamma^\mu + m$ is odd, the relative sign flips, and the anticommutator is required instead. Spin $\frac{1}{2} \Rightarrow$ Fermi statistics $\Rightarrow$ the Pauli exclusion principle.

The summary of this lecture in one sentence. An internal symmetry commuting with Poincaré, realized à la Wigner–Weyl, assigns a conserved charge to each particle; microcausality forces the field to contain both an annihilation and a creation piece; and the only way both pieces can rotate with the same phase is if the second one creates a *distinct* particle of opposite charge and identical mass and spin. Antiparticles are not an assumption — they are a theorem.